

A. Verification verdict

REVISE

B. Brief justification

The architecture is coherent: definitions are sensibly ordered, notation is largely consistent, the timing logic is recoverable, and the deterministic running example works. No circular definitions or undefined internal references were found.

However, the general stochastic system is not yet closed on every reachable history. Several central transitions can become undefined. These are repairable without changing the model's architecture, so the appropriate verdict is REVISE rather than FAIL.

C. Issues required to justify the verdict

1. A restorative refinement need not exist.

The paper asserts that the singleton partition always restores coherence. With stochastic mechanisms, however, repeated observations of the same state and action can produce an empirical conditional outside the test tolerance of every mechanism. No partition of S can separate observations generated at the same state. Consequently,

$$\mathcal{R}(\mathcal{P}_i, r_i) = \emptyset$$

can occur. This conflicts with both the technology's successful-output definition and the assumption that every elected inquiry succeeds with probability at least $\underline{\pi} > 0$.

2. The inquiry-value statistic is not yet a fully defined mathematical object.

"Pinned distinctions" do not formally generate a partition or information algebra, so measurability with respect to them is undefined. Likewise, $B_i(r_i)$ lacks an explicit optimization formula specifying:

- how robustness across rival behaviors is evaluated;
- whether uncertainty over mechanisms is also included;
- how payoffs are normalized when the retained record grows while a generation remains T episodes; and
- how counterfactual payoffs are computed absent the transparency condition, presently stated only in a remark.

Therefore the participation rule $\lambda_i B_i \geq \gamma_i$ cannot yet be evaluated for the general model.

3. The dynamic state and belief transition are not closed.

SCUE is defined using $(\mathcal{P}_i, \mu_i, q_i, \sigma_i)$, but its active-set and inquiry conditions depend on r_i . Different frozen records can therefore give the same listed profile different SCUE status. In addition, if a fluctuating stochastic test empties an active set and later refills it with theories receiving zero mass under the last restricted belief, the prescribed renormalization is undefined. The state must include the record—or a sufficient statistic for it—and the refill rule must guarantee a well-defined belief or explicitly continue the carry-forward policy.